

• Left handed spinor $\psi_\alpha : (\frac{1}{2}, 0)$

→ Lagrangian must be Lorentz invariant and Hermitian
 $\psi\psi + \psi^\dagger\psi^\dagger$

• Need Derivative term:

Guess: $\partial^\mu\psi\partial_\mu\psi$: Hamiltonian is unbounded from below

→ To get a bounded value we must incorporate
 $-(\partial_0\psi)^2 - \partial_\mu\psi\partial_\mu\psi$

• From the previous section we found $\bar{\psi}\bar{\sigma}^\mu\chi$ to be invariant under $\dagger$, so we try:

$$\begin{aligned} (i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\chi)^\dagger &= (i\bar{\psi}_i\bar{\sigma}^{\mu\dot{\alpha}\alpha}\partial_\mu\chi_\alpha)^\dagger \\ &= (-i)(\partial_\mu\chi_\alpha)^\dagger(\bar{\sigma}^{\mu\dot{\alpha}\alpha})^\dagger(\bar{\psi}_i)^\dagger \\ &= -i\partial_\mu\chi_{\dot{\alpha}}\bar{\sigma}^{\mu\dot{\alpha}\alpha}\psi_\alpha \\ &= -i\partial_\mu(\chi_{\dot{\alpha}}\bar{\sigma}^{\mu\dot{\alpha}\alpha}\psi_\alpha) + i\chi_{\dot{\alpha}}\bar{\sigma}^{\mu\dot{\alpha}\alpha}\partial_\mu\psi_\alpha \end{aligned}$$

term would vanish on $\partial\mathcal{M}$

so $(i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\chi)$ is Hermitian and we find

$$\mathcal{L} = i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi - \frac{1}{2}m\psi\psi - \frac{1}{2}m^*\bar{\psi}\bar{\psi} \quad (m \in \mathbb{C})$$

Claim: we can take $m \in \mathbb{R} \rightarrow |m| \in \mathbb{R}$

Proof: $m = |m|e^{i\alpha}$

$$- \frac{1}{2}|m|(e^{i\alpha/2}\psi)(e^{i\alpha/2}\psi)$$

$$\text{define } \hat{\psi} = e^{i\alpha/2}\psi \Rightarrow \hat{\psi}^\dagger = e^{-i\alpha/2}\psi^\dagger$$

$$= -\frac{1}{2}|m|\hat{\psi}\hat{\psi}$$

and so we may take $\mathcal{L} = \mathcal{L}[\hat{\psi}, \hat{\psi}^\dagger]$ and
 $m \rightarrow |m| \in \mathbb{R}$

$$\mathcal{L} = i\bar{\psi}\bar{\sigma}^\mu\partial_\mu\psi - \frac{1}{2}m\psi\psi - \frac{1}{2}m\bar{\psi}\bar{\psi} \quad (m \in \mathbb{R})$$

$$\text{E.O.M: } \frac{\delta\mathcal{L}}{\delta\psi^\dagger} = i\bar{\sigma}^\mu\partial_\mu\psi - \frac{1}{2}m\bar{\psi} = 0$$

$$\text{or } [-i \bar{\sigma}^{\mu\dot{\alpha}\beta} \partial_\mu \psi_\beta + m \bar{\psi}^{\dot{\alpha}} = 0$$

• of this

$$i \bar{\sigma}^{\mu\dot{\alpha}\beta} \partial_\mu \bar{\psi}_{\dot{\beta}} + m \psi^\alpha = 0$$

$$i \varepsilon^{\dot{\beta}\gamma} \sigma^\mu_{\alpha\gamma} \partial_\mu \bar{\psi}_{\dot{\beta}} + m \psi_\alpha = 0$$

$$-i \sigma^\mu_{\alpha\dot{\gamma}} \partial_\mu \bar{\psi}^{\dot{\gamma}} + m \psi_\alpha = 0$$

• Combine these into a single expression

$$\begin{pmatrix} m \delta^\alpha_\gamma & -i \sigma^\mu_{\alpha\dot{\gamma}} \partial_\mu \\ -i \bar{\sigma}^{\mu\dot{\alpha}\gamma} \partial_\mu & m \delta^{\dot{\alpha}}_{\dot{\gamma}} \end{pmatrix} \begin{pmatrix} \psi_\gamma \\ \bar{\psi}_{\dot{\gamma}} \end{pmatrix} = 0$$

$$\Rightarrow \left[m \begin{pmatrix} \delta^\alpha_\gamma & 0 \\ 0 & \delta^{\dot{\alpha}}_{\dot{\gamma}} \end{pmatrix} - i \begin{pmatrix} 0 & \sigma^\mu_{\alpha\dot{\gamma}} \\ \bar{\sigma}^{\mu\dot{\alpha}\gamma} & 0 \end{pmatrix} \partial_\mu \right] \begin{pmatrix} \psi_\gamma \\ \bar{\psi}_{\dot{\gamma}} \end{pmatrix} = 0$$

We now define two objects:

(i) Gamma matrices: $\gamma^\mu: \{\gamma^\mu, \gamma^\nu\} = -2\eta^{\mu\nu}$

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu_{\alpha\dot{\gamma}} \\ \bar{\sigma}^{\mu\dot{\alpha}\gamma} & 0 \end{pmatrix}$$

(ii) Majorana field:

$$\Psi = \begin{pmatrix} \psi_\gamma \\ \bar{\psi}_{\dot{\gamma}} \end{pmatrix}$$

• with $\{\sigma^\mu, \bar{\sigma}^\nu\}_\alpha = -2\eta^{\mu\nu} \delta_\alpha^\gamma$, $\{\bar{\sigma}^\mu, \sigma^\nu\}_{\dot{\alpha}} = -2\eta^{\mu\nu} \delta_{\dot{\alpha}}^{\dot{\gamma}}$
 $\{\gamma^\mu, \gamma^\nu\} = -2\eta^{\mu\nu}$

$$(m \mathbb{1} - i \gamma^\mu \partial_\mu) \Psi = 0$$

(Dirac Equation)

* Consider a theory w two left-handed spinor fields with an $SO(2)$ symmetry

$$\mathcal{L} = i \psi_i^\dagger \bar{\sigma}^\mu \partial_\mu \psi_i - \frac{1}{2} m \psi_i \psi_i - \frac{1}{2} m \psi_i^\dagger \psi_i^\dagger$$

$i=1, 2$ (Sum over i)

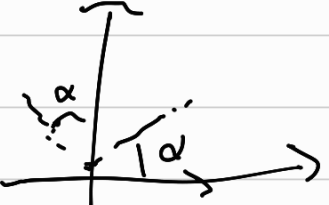
• $SO(2)$ symmetry

$$\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \rightarrow \begin{pmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$$

$$\text{or } \psi_1' = \cos(\alpha) \psi_1 + \sin(\alpha) \psi_2$$

$$\psi_2' = -\sin(\alpha) \psi_1 + \cos(\alpha) \psi_2$$

Note $SO(2) \cong U(1)$ since



$$\begin{aligned}
 x &\rightarrow x \cos(\alpha) + y \sin(\alpha) = x' \\
 y &\rightarrow -x \sin(\alpha) + y \cos(\alpha) = y'
 \end{aligned}$$

However in $\mathbb{C}$ $z \equiv x + iy = |r|e^{i\theta}$

$$z \rightarrow z' = z e^{-i\alpha}$$

$$(x + iy)(\cos(\alpha) - i \sin(\alpha))$$

$$\begin{aligned}
 &x \cos(\alpha) - i x \sin(\alpha) + y \sin(\alpha) + i y \cos(\alpha) \\
 &= (x \cos(\alpha) + y \sin(\alpha)) + i (y \cos(\alpha) - x \sin(\alpha)) \\
 &\quad \tilde{x} + i \tilde{y}
 \end{aligned}$$

• So we have $SO(2) \cong U(1)$

* We have

$$\chi \equiv \frac{1}{\sqrt{2}}(\psi_1 + i\psi_2) \quad \psi_1 = \frac{1}{\sqrt{2}}(\chi + \xi)$$

$$\xi \equiv \frac{1}{\sqrt{2}}(\psi_1 - i\psi_2) \quad \psi_2 = \frac{1}{\sqrt{2}i}(\chi - \xi)$$

$$\frac{1}{2}i(\chi + \xi)^\dagger \bar{\sigma}_\mu (\chi + \xi) - \frac{1}{2}m(\frac{1}{\sqrt{2}}(\chi + \xi) | \frac{1}{\sqrt{2}}(\chi + \xi) |)$$

... typo in Eq. (36.16)

$$\mathcal{L} = i\chi^\dagger \bar{\sigma}^\mu \partial_\mu \chi + i\xi^\dagger \bar{\sigma}^\mu \partial_\mu \xi - m\chi\xi - m\xi^\dagger \chi^\dagger$$

$U(1)$ -symmetry

$$\chi \rightarrow e^{-i\alpha}\chi$$

$$\xi \rightarrow e^{i\alpha}\xi$$

E.o.M of ⊗
$$\begin{pmatrix} m\delta_a^c & -i\sigma_{ac}^\mu \partial_\mu \\ -i\sigma^{a\dot{c}} \partial_\mu & m\delta_{\dot{c}}^{\dot{a}} \end{pmatrix} \begin{pmatrix} \chi_c \\ \xi^{\dagger \dot{c}} \end{pmatrix} = 0$$

Define a Dirac field

$$\Psi_D \equiv \begin{pmatrix} \chi_c \\ \xi^{\dagger \dot{c}} \end{pmatrix}$$

• Exactly the same solution to the Dirac Equation. Take

$$\bar{\Psi}_D^\dagger = (\chi^\dagger_{\dot{a}}, \xi^c)$$

Introduce the matrix

$$B \equiv \begin{pmatrix} 0 & \delta_{\dot{a}}^c \\ \delta_a^{\dot{c}} & 0 \end{pmatrix}$$

$B \equiv \gamma^0$ Numerically but their index structure is important

Dirac Conjugate:

$$\bar{\Psi}_D = \Psi_D^\dagger \beta$$

$$\bar{\Psi}_D = (\chi_a^\dagger, \xi^c) \begin{pmatrix} 0 & \sigma^0_{\dot{a}c} \\ \sigma^0_{c\dot{a}} & 0 \end{pmatrix} = (\xi^{\dot{a}}, \chi_a^\dagger)$$

* Thus, $\bar{\Psi}_D \Psi_D = (\xi^{\dot{a}}, \chi_a^\dagger) \begin{pmatrix} \chi^{\dot{a}} \\ \xi^{\dagger \dot{a}} \end{pmatrix}$
 $= \xi^{\dot{a}} \chi_{\dot{a}} + \chi_a^\dagger \xi^{\dagger \dot{a}}$

=> Now, consider

$$\bar{\Psi}_D \gamma^\mu \partial_\mu \Psi_D$$

$$= (\xi^{\dot{a}}, \chi_a^\dagger) \begin{pmatrix} 0 & \sigma^\mu_{\dot{a}c} \\ \bar{\sigma}^{\mu\dot{a}c} & 0 \end{pmatrix} \partial_\mu \begin{pmatrix} \chi_c \\ \xi^{\dagger \dot{c}} \end{pmatrix}$$

$$= (\xi^{\dot{a}}, \chi_a^\dagger) \begin{pmatrix} \sigma^\mu_{\dot{a}c} \partial_\mu \xi^{\dagger \dot{c}} \\ \bar{\sigma}^{\mu\dot{a}c} \partial_\mu \chi_c \end{pmatrix}$$

$$= \xi^{\dot{a}} \sigma^\mu_{\dot{a}c} \partial_\mu \xi^{\dagger \dot{c}} + \chi_a^\dagger \bar{\sigma}^{\mu\dot{a}c} \partial_\mu \chi_c$$

$$\hookrightarrow \xi^{\dot{a}} \sigma^\mu_{\dot{a}c} \partial_\mu \xi^{\dagger \dot{c}} = -\partial_\mu \xi^{\dot{a}} \sigma^\mu_{\dot{a}c} \xi^{\dagger \dot{c}} + \partial_\mu (\xi^{\dot{a}} \sigma^\mu_{\dot{a}c} \xi^{\dagger \dot{c}})$$

• If we ignore the Boundary term

$$\bar{\Psi}_D \gamma^\mu \partial_\mu \Psi_D = \xi^{\dagger \dot{a}} \bar{\sigma}^{\mu\dot{a}c} \partial_\mu \xi_c + \chi_a^\dagger \bar{\sigma}^{\mu\dot{a}c} \partial_\mu \chi_c$$

- Used anti commutativity and flipped $\dot{a} \dot{c} \rightarrow \dot{c} \dot{a}$
 which gives two (-1) values $\alpha_a \rightarrow \alpha_a$

$$\mathcal{L} = i \bar{\Psi}_D \gamma^\mu \partial_\mu \Psi_D - m \bar{\Psi}_D \Psi_D$$

which has a U(1)-symmetry $e^{-i\alpha} \Psi_D$

Noether current: $J^\mu = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \Psi_D)} \delta \Psi_D$

Here $\Psi_D \rightarrow \Psi_D + \delta \Psi_D$

$$J^\mu = i \bar{\Psi}_D \gamma^\mu (-i\alpha \Psi_D) = \alpha \bar{\Psi}_D \gamma^\mu \Psi_D$$

$$= \alpha (\chi^\dagger \bar{\sigma}^\mu \chi - \xi^\dagger \bar{\sigma}^\mu \xi)$$

* Charge Conjugation: Discrete symm. which enlarges $SO(2) \rightarrow O(2)$.

(charge conj) $\chi \longleftrightarrow \xi$

Define the charge conj. operator C

$$\begin{aligned} C^{-1} \chi_\alpha(x) C &= \xi_\alpha(x) \\ C^{-1} \xi_\alpha(x) C &= \chi_\alpha(x) \end{aligned}$$

* Charge Conjugation matrix

want to define $\mathcal{C}$ to be even so that

$$C^{-1} \mathcal{C} C = \mathcal{C}$$

$$\Phi_D^T = \begin{pmatrix} \xi^\alpha \\ \chi^\dagger_i \end{pmatrix} = \Phi_D^* \beta^*$$

then

$$\Phi_D^C = C \Phi_D^T = \begin{pmatrix} \epsilon_{\alpha\gamma} & 0 \\ 0 & \epsilon_{ij} \end{pmatrix} \begin{pmatrix} \xi^\alpha \\ \chi^\dagger_i \end{pmatrix} = \begin{pmatrix} \xi_\gamma \\ \chi^\dagger_j \end{pmatrix}$$

(charge conjugate)

• Thus Φ_D^C is Φ_D with the spins flipped

$$\chi \quad C^{-1} \Phi_D C = \Phi_D^C$$

• Charge conj. Matrix satisfies $C = \begin{pmatrix} -\epsilon^{\alpha\gamma} & 0 \\ 0 & -\epsilon_{ij} \end{pmatrix}$

$$C^T = C^\dagger = C^{-1} = -C$$

$$C^{-1} \gamma^\mu C = \begin{pmatrix} 0 & -\bar{\sigma}^{\mu a \dot{a}} \\ -\sigma^{\mu}_{i \dot{i}} & 0 \end{pmatrix}$$

and so $C^{-1} \gamma^\mu C = -(\gamma^\mu)^T$

* For a Majorana field $\Phi_M^C = \Phi_M$

$$\Phi_M^C = \Phi_M \quad \leftrightarrow \quad \psi^\dagger = \psi \quad \text{IR}$$

$$\Phi_D^C = C^{-1} \Phi_D C \quad \leftrightarrow \quad \psi \in \mathbb{R}$$

* Thus, for a Majorana field $\Phi_M = \begin{pmatrix} \psi_a \\ \psi^\dagger_{\dot{a}} \end{pmatrix}$

(Majorana)

$$\mathcal{L} = \frac{i}{2} \bar{\Psi}_\mu \not{\partial}_\mu \Psi_\mu - \frac{1}{2} m \bar{\Psi}_\mu \Psi_\mu$$

→ Does not include $\Psi^c = \Psi$ condition

$$\left. \begin{aligned} \Psi_\mu &= C \bar{\Psi}_\mu^T \\ \bar{\Psi}_\mu^T &= \bar{\Psi}_\mu C^{-1} \end{aligned} \right\} \bar{\Psi}_\mu = \Psi_\mu^T C$$

$$\mathcal{L} = \frac{i}{2} \bar{\Psi}_\mu \not{\partial}_\mu \Psi_\mu - \frac{1}{2} m \bar{\Psi}_\mu \Psi_\mu$$

$$\gamma_5 \equiv \begin{pmatrix} -\delta_a^b & 0 \\ 0 & +\delta_{\dot{a}}^{\dot{b}} \end{pmatrix}$$

$$\begin{aligned} P_L &\equiv \frac{1}{2}(1 - \gamma_5) = \begin{pmatrix} \delta_a^c & 0 \\ 0 & 0 \end{pmatrix} \\ P_R &\equiv \frac{1}{2}(1 + \gamma_5) = \begin{pmatrix} 0 & 0 \\ 0 & \delta_{\dot{a}}^{\dot{b}} \end{pmatrix} \end{aligned}$$

$$P_L \Psi_0 = \begin{pmatrix} \chi_c \\ 0 \end{pmatrix}, \quad P_R \bar{\Psi}_0 = \begin{pmatrix} 0 \\ \bar{\xi}^{\dot{a}} \end{pmatrix}$$

$$\gamma_5 = i \gamma_0 \gamma^1 \gamma^2 \gamma^3$$

$$= -\frac{i}{24} \epsilon_{\mu\nu\rho\sigma} \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma$$

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu] = \begin{pmatrix} + (S_L^{\mu\nu})_a^b & 0 \\ 0 & - (S_R^{\mu\nu})_{\dot{a}}^{\dot{b}} \end{pmatrix}$$

$$U(\Lambda)^{-1} \bar{\Psi}_{D,\mu} U(\Lambda) = D(\Lambda) \bar{\Psi}_{D,\mu} (\Lambda^{-1} x)$$

$$D(1 + \delta\omega) = 1 + \frac{i}{2} \delta\omega_{\mu\nu} S^{\mu\nu}$$